Python computing lab course

Engineering Communication, Measurement, and Data Analysis

Learning the basics of Python

Objectives

After completing these exercises you will be able to:

- Create Jupyter notebooks with cells containing Python code
- Create and be able to identify variables of different types
- Perform simple calculations using Python

Introduction

These computer lab sessions will introduce Python through Jupyter Notebooks, they will build on the content covered in pre-session notebooks. It is important that you have gone through and completed the mini-tasks within those notebooks before attempting these exercises. For each session create a new notebook with a reasonable name.

Getting help

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There are number of different resources you can use if you get stuck and need help.

1. In case of an error, read the compiler error. It will often tell you the line where the error is occurring. Try to correct the error and re-run the compiler.
2. Still getting an error, try copying the error message into google, or search using some keywords.
3. Still a bit stuck, ask an demonstrator for help, who will be able to point you in the right direction

You can also refer back to the pre-lab material and the quick reference sheet on blackboard.

Please note that Exercises 1 to 6 are designed to develop the skills required for the successful completion of this unit. These are **core** exercises. Exercises 7 and 8 extend beyond the level necessary for this unit and introduce skills that will be beneficial later in the programme. These are **extension** exercises.

1 Exercises

Exercise 1 (core): Strings

Run the following piece of code in the Jupyter notebook:

```
x = ‘‘Chocolate Cake’’
```

For each of the commands below, record the output and what the program has done in the notes section.

Command	Output	Notes
`x[0]`	`C`	found the first letter of string x
`x[-1]`		
`len(x)`		
`x[1:3]`		
`x[:5]`		
`x.upper()`		

Exercise 2 (core): Data Types

Identify the type of data for each variable and convert (cast) to the type shown in the table.

Variable	Data Type	Cast Type	Output	Notes
`3.4`	Float	Integer	3	Outputs the integer part of the number
`True`		Float		
`False`		String		
`3`		Boolean		
`5.1`		Boolean		
`7`		Float		

Exercise 3 (core): Simple Calculations and Comparison Operators

Create and assign the following variables to use in your calculations:

- `x = 2`
- `y = 5.7`
- `z = -8`

Command	Output	Notes
`x + 2 * y`	`13.4`	
`z % x`		
`y ** x`		
`z - True`		
`z >= x`		
`y != x`		
`x >= 1 and y == 6`		
`y < 10 or z > 1`		
`a = y >= 5`		

Exercise 4 (core): Outputting a String

Write a short piece of code that calculates the remainder from the variable `b` (integer value) when it is divided by two and outputs the following string:

“The remainder when (value of b) is divided by 2 is (value of remainder).”

Example output:

Exercise 5 (core): Float Formatting

Format the value of `E` into exponent notation (scientific notation) to 3 decimal places.

```
E = 197931859
```

Output:

Exercise 6 (core): Young's Modulus

The modulus of elasticity (Young's Modulus, E) of a material can be calculated from experimental results. The Young's Modulus of a material can be calculated by dividing the stress by strain, Equation 1. Stress is the force applied per unit area and therefore has units of force per unit area, typically Nm^{-2} but engineers often use areas square millimetres leading to units of N mm^{-2} or MPa, Equation 2. Strain is the relative change in length (tension/compression) or the angle of twist (shear). Strain is dimensionless and has no units, Equation 3.

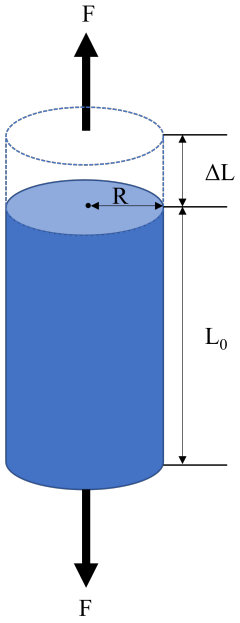


Figure 1: Cylindrical rod in tension

$$E = \sigma/\epsilon \tag{1}$$

$$\sigma = F/A \tag{2}$$

$$\epsilon = \Delta L/L_0 \tag{3}$$

Where E = Young's modulus (MPa or GPa) , σ = Stress (N mm^{-2} or MPa), ϵ = Strain (no units), F = Force (N), A = cross-sectional area (mm2), ΔL = change in length (L-L_0, mm); where L = final length (mm) and L_0=initial length (mm).

A series of experiments were conducted using the test setup in Figure 1, each test used a different material with different starting dimensions (cross-sectional area and initial length) as shown in Table 1. Using the equations given, the predicted change in and final length of the rods can be calculated. Note: Ensure that consistent units are used throughout calculations (units do not need to be taken to base SI units) and MPa = Nmm^{-2} should be used. The value of 3.14 should be used for π, this will need included in your code as a variable.

Material	Young's Modulus (GPa)	Force (N)	Radius (mm)	Initial length (mm)
Steel	200	8000	1.1	500
Aluminium	60	5000	2.5	750
Titanium	115	4000	3	1000

Table 1: Table of material properties and dimensions

Task 6
Write a program that calculates and outputs the change in and final length in mm of each sample in Table 1. Record the results in the table below or in your lab notebook.

	Change in length (mm)	Final length (mm)
Steel sample		
Aluminium sample		
Titanium sample		

Hint: 1 GPa = 1000 MPa. Use the following units MPa (Nmm^{-2}), N and mm. By using the Young's modulus of each material and your program to determine which of the materials should be selected to fulfil the following design criteria. A material that will have a change in length of 2.5 mm from its original length of 200mm and radius of 1mm, when a load of 4500N is applied. (You do not need to re-write your code just change the variable values).

Answer:

Exercise 7 (extension): Evaluating the lift of an aerofoil

The lift produced by an aerofoil can be determined from the lift coefficient and the dynamic pressure on the aerofoil, see equation 4. The lift (l) and drag (d) are affected by the angle of attack (α), air density (ρ) and freestream velocity (ν), see Figure 2. The lift (per unit chord) produced by an

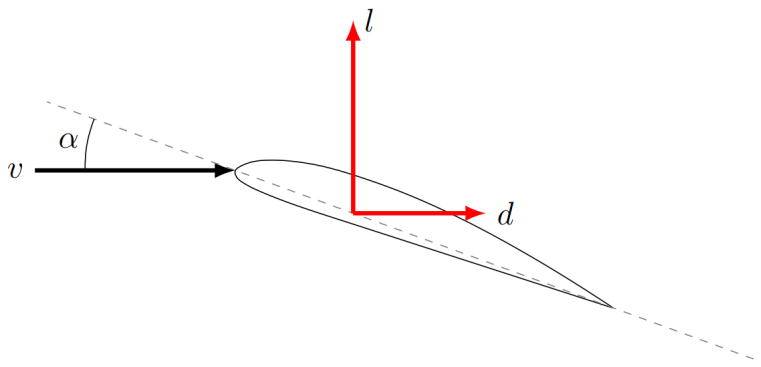


Figure 2: Forces acting on an aerofoil

aerofoil in inviscid flow can be approximated as:

$$l = qC_l \tag{4}$$

where $q = \frac{1}{2}\rho\nu^2$ is the dynamic pressure (ρ is the air density and ν is the freestream velocity) and $C_l = 2\pi\alpha$ is the lift coefficient (α is the angle of attack into the flow direction in radians).

Task 7
Write a program that determines the lift force in N for the following variable values:
$\nu = 10m/s$, $\rho = 1.225kg/m^3$ and $\alpha = 10°$. The program should output a string that says "The lift is (value of l) N" with lift being given to 3 decimal places.

Answer:

Exercise 8 (extension): Moment of inertia

The mass of a simple pendulum is concentrated at a single point at a distance from the pivot, while a compound pendulum's mass is not a fixed point but a rigid body, see Figure 3. The period of the

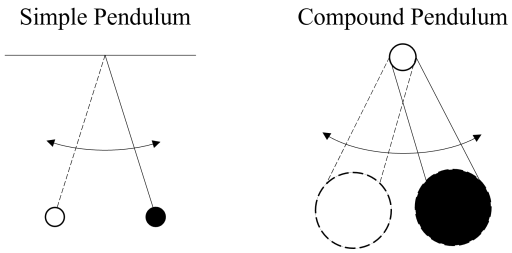


Figure 3: Diagram of a simple and compound pendulum

compound pendulum (T) depends on the properties of the body: its mass (m), moment of inertia (I) and the distance from the pivot point to the centre of mass (d), as shown in Equation 5.

$$T = 2\pi\sqrt{\frac{I + md^2}{mgd}} \tag{5}$$

Where T = period of oscillation (s), I = moment of inertia (kg m^2), m = mass of rigid body (kg), d = distance from the pivot to the centre of mass of the body (m) and g = acceleration due to gravity (9.81 ms^{-2}).

An experiment was ran to determine the period of oscillation for Object A, to improve accuracy the time period was measure 5 times and measured for 10 oscillations. The results of the experiment are shown in Table 2.

Experiment	**1**	**2**	**3**	**4**	**5**
Time period for 10 oscillations (s)	11	12	11	12	9

Table 2: Time period for 10 oscillations for Object A

Task 8
Write a program that determines the moment of interia for Object A. Object A has m = 0.5kg and d = 0.14 m. The program should print a string that contains the value of I to 3 decimal places and in scientific notation with units as its output.

Answer: